

Seong-Jun Kang, Ph.D.

Immunologist working on single-cell and multiplexed measurement of immune responses, in transplantation, neuroinflammation, and inflammatory disease

Postdoctoral Fellow applicant · BioTranslation Lab, Center for Nanomedicine, Brigham and Women's Hospital and Harvard Medical School (Position 3: Immunology and Immunoengineering)

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Available August 2026

RESEARCH PROFILE

I am an immunologist, and I measure what the immune system does after an intervention. During my doctoral training I worked in a laboratory that ran preclinical islet transplantation in non-human primates, and eight of my publications come from that work, on the systemic side of immunosuppression. At PB Immune Therapeutics I led an immune-regulatory antibody program, built the in-house single-cell and spatial analysis capability, and designed and ran EAE studies in cynomolgus monkeys and marmosets. In transplantation the question I have worked on is how narrow the suppression can be and still keep the graft. Materials and drug delivery would be new to me.

POSITION-RELEVANT EXPERTISE

- **Immunosuppression for transplantation.** Co-author on non-human-primate islet transplantation studies of **anti-CD40 (2C10R4)** regimens, of a **JAK inhibitor used in place of tacrolimus**, and of **IL-6 blockade**. Co-first author on long-term diabetes control with a tofacitinib-based regimen in rhesus monkeys, and a co-inventor on a humanized anti-CD40 antibody patent from 2024.
- **Immune readout after an intervention.** High-dimensional flow cytometry with panels up to 24 markers, designed in house; 10-color six-way sorting; **Olink** targeted plasma proteomics on the immune and neurology panels; longitudinal immune monitoring so the compartment is read over time rather than at a single endpoint.
- **Neuroinflammation, measured end to end.** Whole-brain immune cell isolation through to single-cell analysis run myself: enzymatic digestion, Percoll gradient, FACS, 10x, and the analysis. Ten immune populations from 11,587 cells in an Alzheimer's disease mouse model, with exhausted CD8⁺ tissue-resident memory T cells interacting with choroid-plexus epithelium at the blood-CSF barrier.
- **Single-cell and spatial analysis at scale.** scRNA-seq, scTCR-seq, snATAC-seq, **GeoMx DSP** and **Visium**, at the bench and in code. Cell subsets decomposed at 251,055-cell scale with **consensus NMF**, cohorts integrated across more than 700 donors, 8 organs and 23 immune diseases, and the result checked with held-out AUC and calibration.
- **Studies that arrive in cohorts.** Reference material in every collection round and batch written into the model as a variable. In nested designs I test at the level of the donor or the animal rather than the cell, and the same arithmetic applies when a study has six animals per arm.
- **Program execution and grant writing.** External studies designed and placed, assay conditions specified to the vendor, milestones tracked and advancement decided against fixed dates. About \$1.3 million across five industry proposals, lead writer on four, always under the institutional PI and not as PI. K99/R00 and LSRF are open to me, and I would apply within the first two years.

EDUCATION

DEGREES

Ph.D.
2014-2023 **Seoul National University**, Seoul, Republic of Korea
Department of Biomedical Sciences
Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts
Advisor: Chung-Gyu Park, M.D., Ph.D.

M.S.
2012-2014 **Seoul National University**, Seoul, Republic of Korea
Department of Biomedical Sciences
Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14
Advisor: Chung-Gyu Park, M.D., Ph.D.

B.S. Sungkyunkwan University, Suwon, Republic of Korea
2008–2012 Department of Genetic Engineering

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.
Associate Director 2023–2025	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Led a humanized anti-CD40 antibody program: candidate ranking by surface plasmon resonance (Biacore) and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.
Manager 2021–2023	PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.
Research Student 2020–2023	SCAID, Single Cell Atlas of Immune Diseases, Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.
Researcher 2019–2021	Xenotransplantation Research Center, Seoul National University, Seoul, Republic of Korea Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.
Teaching Assistant 2014–2016	Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

SELECTED RESEARCH OUTPUTS

- 1 Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation* 2024. [Download PDF](#)
How narrow the suppression can be and still keep the graft. Co-first author on long-term diabetes control with a tofacitinib-based immunosuppressive regimen in rhesus monkeys, a JAK inhibitor standing in for calcineurin blockade. Tissue processing and longitudinal immune monitoring on the study.
- 2 Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." *Xenotransplantation* 2018.
Costimulation blockade in a preclinical xenotransplantation model. Pig-to-non-human-primate islet transplantation under an anti-CD40 (2C10R4) regimen. My work was on the immune monitoring side, tracking what the regimen did to the recipient compartment rather than graft survival alone.

- 3 **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology* 2025. [Download PDF](#)

Immune entry into the brain, measured from tissue to cell state. Whole-brain harvest, enzymatic digestion, Percoll enrichment, CD45⁺ FACS, 10x, then the analysis, all run by me. 11,587 cells and ten immune populations; exhausted CD8⁺ tissue-resident memory T cells engaging choroid-plexus epithelium, with MHC class I, interferon-stimulated genes and JAK-STAT raised on that epithelium.

- 4 Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology* 2026. in revision

A phenotype held to a batch-aware design, then tested. Four matched cohorts, 16 donors each, pooled four per lane and demultiplexed by SNP so donor and sequencing run were not confounded. 251,055 cells; custom UMI-aware, BAM-level code resolving PDE4D isoform identity from 5' reads; consensus NMF programs repeating across 702 samples; held-out AUC 0.780 for arthritis. Short-isoform overexpression raised the pathogenic fraction and a selective inhibitor lowered it.

MENTORING AND RESEARCH TRAINING

2013 **Sung Ha Lim, M.D., Ph.D.** STUDENT INTERN

Postdoctoral Fellow; Laboratory of Shadmehr Demehri, M.D., Ph.D.; Department of Dermatology, Massachusetts General Hospital; Harvard Medical School, Boston, MA, USA

Training: Cell culture, Western blotting, flow cytometry, and C57BL/6 mouse breeding.

2015 **Soyoung Jeong, Ph.D.** STUDENT INTERN

Postdoctoral associate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: Cell culture, flow cytometry, and in vivo mouse experiments.

2016 **Sowon Choi, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: Mouse pancreatic islet isolation, pancreatic islet beta-cell functional assays, and ion channel assays.

2021 **Hyo Jeong Nam, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea; Visiting Scholar; Laboratory of Seunghee Kim-Schulze, Ph.D.; Department of Immunology and Immunotherapy; Icahn School of Medicine at Mount Sinai, New York, NY, USA

Training: GeoMx DSP, T-cell functional assays, and single-cell RNA sequencing (scRNA-seq) data analysis using R.

2021 **Beom Keun Cho, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: GeoMx DSP, mouse breeding, cell culture, T-cell functional assays, and scRNA-seq data analysis using R.

2021 **Minji Kang, Ph.D.** INCOMING GRADUATE STUDENT

Postdoctoral associate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: GeoMx DSP spatial transcriptomics data analysis and T-cell functional assays.

2021 **Jung Ho Lee, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate, Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea; Visiting Scholar, Laboratory of Ulrich von Andrian, M.D., Ph.D.; Department of Immunology, Harvard Medical School, Boston, MA, USA

Training: Analysis of scRNA-seq data using R.

2021 **Hyun Seung Choi, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: Analysis of scRNA-seq data using R.

2022 **Yuji Lee, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.

2022 **Sanha Hwang, M.S.** INCOMING GRADUATE STUDENT

Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea

Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.

2022	Dongjun Kim, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.	2022	Yewon Moon, M.S. INCOMING GRADUATE STUDENT Ph.D. Candidate; Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Regulatory T-cell suppression assays and scRNA-seq data analysis using R and machine-learning models.
2022	Karoliina Eliisa Ruhno, M.S. INCOMING GRADUATE STUDENT Senior Researcher; IMBiologics Corp., Suwon, Republic of Korea Training: Regulatory T-cell suppression assays.	2023	Moongi Chae, M.S. STUDENT INTERN Ph.D. Student; Johns Hopkins University, Baltimore, MD, USA Training: Immune-cell culture, flow cytometry, and scRNA-seq data analysis using machine-learning models and AI tools.
2023	Sung Bin Bae, M.S. INCOMING GRADUATE STUDENT Department of Biomedical Sciences; Seoul National University Graduate School, Seoul, Republic of Korea Training: Cell culture, flow cytometry, and GeoMx DSP spatial-transcriptomics data analysis.	2024	Mi Jeong Oh, M.S. MANAGER, PB IMMUNE THERAPEUTICS Manager; PB Immune Therapeutics, Seoul, Republic of Korea Training: Experimental design for animal sample collection, ELISA and SPR assay design, and 10x Genomics scRNA-seq.
2024	Eun Sol Jo, B.S. ASSISTANT MANAGER, PB IMMUNE THERAPEUTICS Assistant Manager; PB Immune Therapeutics, Seoul, Republic of Korea Training: Animal experiment design and evaluation of a mouse model of experimental autoimmune encephalomyelitis (EAE).		

TECHNICAL EXPERTISE			
	Transplantation & in vivo immunology	Non-human-primate islet transplantation, systemic immunosuppression side: co-author on studies of anti-CD40 (2C10R4) , of a JAK inhibitor used in place of tacrolimus , and of IL-6 blockade ; co-first author on long-term diabetes control with a tofacitinib-based regimen in rhesus monkeys (<i>Xenotransplantation</i> 2024). Tissue processing, longitudinal immune monitoring, and brain and spinal cord harvest on those studies. EAE studies in cynomolgus monkeys and marmosets designed and run for multiple sclerosis programs. Mouse Alzheimer's disease neuroimmune profiling: whole-brain digestion, Percoll gradient, CD45 ⁺ FACS, 10x, analysis (<i>J Neuroimmunol</i> 2025, co-first author).	
	Cell, tissue & immunological assays	Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels).	
	Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and 10x Chromium Connect automated preparation. Seurat , Scanpy , Squidpy , Monocle3 , CellChat/NicheNet ; scVI/scANVI, batch integration, consensus NMF ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.	
	Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.	
	Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.	

Program & external execution	CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection.
Binding & protein interaction	Surface plasmon resonance (Biacore) run hands-on for antibody lead work: association and dissociation rates, affinity ranking across a candidate panel, receptor-blocking confirmation, and human / non-human-primate cross-reactivity. ELISA-based affinity and epitope analysis; competition binding. Antibody lead selection and optimization from candidate panel to registered patent. Functional validation of a peptide inhibitor of calcineurin in primary human T cells (CABIN1 core peptide, <i>Exp Mol Med</i> 2022).

18

TOTAL PAPERS

17

PEER-REVIEWED

1

IN REVISION

6

FIRST / CO-FIRST


[Google Scholar · Full Profile, Citations & h-index ↗](#)

PUBLICATIONS · FIRST / CO-FIRST AUTHOR

2026

Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." *Nature Immunology*. in revision

2025

Seong-Jun Kang*, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8+ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. [Download PDF](#)

2025

Seong-Jun Kang*, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. [Download PDF](#)
 Co-first author. Single-cell and functional dissection of skin-homing regulatory T cells in atopic dermatitis, with OX40 up-regulated alongside impaired suppressive function.

2024

Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. [Download PDF](#)

2022

Youngkyoung Lim*, **Seong-Jun Kang***, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. [Download PDF](#)
 Co-first author. Ran GeoMx DSP end to end on a nevus-arising melanoma: immunofluorescence staining, fluorescence-guided ROI selection, instrument operation, and the spatial-transcriptomic analysis across 16 depth-segmented regions.

2019

Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, **Seong-Jun Kang***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. [Download PDF](#)
 Co-first author, one of four authors contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

2026	Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, Seong-Jun Kang , Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." <i>Allergy</i> .
2024	Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, Seong-Jun Kang , Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." <i>eBioMedicine</i> . Download PDF
2024	Youngkyoung Lim*, Beom Keun Cho*, Seong-Jun Kang , Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." <i>JEADV</i> . Download PDF
2023	Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORyt through posttranslational modification." <i>Experimental & Molecular Medicine</i> .
2022	Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, Seong-Jun Kang , Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." <i>Experimental & Molecular Medicine</i> .
2021	So Hee Hong, Hyun Je Kim, Seong-Jun Kang , Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." <i>Current Diabetes Reports</i> . Download PDF
2020	Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." <i>Immunology Letters</i> .
2019	Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bonggi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, Seong-Jun Kang , Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." <i>Scientific Reports</i> .
2018	Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, Seong-Jun Kang , Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." <i>Xenotransplantation</i> .
2018	Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, Seong-Jun Kang , Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." <i>Xenotransplantation</i> .
2018	Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, Seong-Jun Kang , Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." <i>Transplantation</i> .
2018	Jung-Sik Kim, Hyunwoo Chung, Nari Byun, Seong-Jun Kang , Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." <i>Biochemical and Biophysical Research Communications</i> .

2026

Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." In preparation

Co-author (immunology). Ran the Olink plasma proteomics across 12 patients and 6 controls and the flow-cytometry immunophenotyping of lymph-node leukocytes.

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SELECTED GRANT WRITING & PROGRAM FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER → ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Anti-CD40 antibody lead selection and optimization (multiple sclerosis)	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris)	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher
NAVER Corporation	Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT)	Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9-22-1-05)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, **not as PI**. Budgets converted at ≈ ₩1,400 / USD.

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AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

2024	Antagonistic Anti-CD40 Humanized Antibody. Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.
2019	Immunosuppression Composition Comprising JAK Inhibitor. Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS

- | | |
|------|---|
| 2024 | Grand Prize, 2024 GPTers AI Hackathon (Project GAIA Team) |
| 2023 | Grand Prize, SBA Smart Workathon (Work Process Innovation) |
| 2017 | Best Poster Award, American Transplant Congress, Chicago · Best Presentation Award, Korean Association of Immunologists |

SCHOLARSHIPS

- | | |
|-----------|---|
| 2014–2020 | NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship |
|-----------|---|

■ SELECTED PRESENTATIONS

- | | |
|--------|---|
| Poster | KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012) |
|--------|---|

■ REFERENCES

Chung–Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology / Biomedical Sciences, SNU College of Medicine; CEO/CSO, PB Immune Therapeutics · chgpark@snu.ac.kr

Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.

Seung–Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global Medical Science, Yonsei University Wonju College of Medicine · skcha@yonsei.ac.kr

Most recent research supervisor (2025–2026).

Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology & Immunology / Biomedical Sciences; Interdisciplinary Program in Artificial Intelligence (IPAI), SNU · tte9801@snu.ac.kr

Co-corresponding on the single-cell studies; supervised the computational work.

Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of Medical Science & Engineering, KAIST, Daejeon · chemami@kaist.ac.kr

Corresponding author, PDE4D / Nature Immunology study.

Seong–Jun Kang · Curriculum Vitae Available from August 2026 · based in Boston, MA · currently authorized to work in the United States without employer sponsorship to begin employment

· BioTranslation Lab

Updated
September
2026